Conserved mitochondrial suppression and organelle-specific transcriptomic responses to spaceflight across six eukaryotic species

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Abstract

Spaceflight induces transcriptomic changes across organisms, but the extent to which subcellular and metabolic pathways are conserved across species remains poorly understood. We performed a cross-species meta-analysis of 22 transcriptomic datasets from the NASA Open Science Data Repository spanning six eukaryotic species (*Homo sapiens*, *Mus musculus*, *Drosophila melanogaster*, *Caenorhabditis elegans*, *Saccharomyces cerevisiae*, and *Arabidopsis thaliana*), encompassing 20,333 differentially expressed genes (DEGs). Using a human-anchored orthology matrix built from OrthoDB v12 (18,030 genes, 651 with orthologs in all five other species), we mapped DEGs to 16 subcellular compartments and applied Fisher's combined probability test to identify conserved organelle-level responses. We report a striking conserved downregulation of mitochondrial genes: 49 orthologs show significant, direction-consistent suppression across at least three species (Fisher FDR $< \mathbf{0.05}$, minimum $\boldsymbol{p = 7.7 \times 10^{-68}}$), including components of Complex III (CYC1), Complex IV (COX5B, COX6A2, COX6B2), ATP synthase (ATP5F1A, ATP5PD), and the TCA cycle (IDH3A, IDH3B, IDH3G). In contrast, peroxisome, lipid droplet, and Golgi apparatus genes show conserved upregulation. KEGG pathway visualization with per-species log2FC overlays reveals that NAD^+-dependent oxidative phosphorylation enzymes are divergently regulated (down in fly, up in human and mouse), while CoA-dependent TCA cycle enzymes are broadly conserved. These findings identify mitochondrial energy metabolism as a deeply conserved target of the spaceflight transcriptional response across a billion years of eukaryotic evolution.

Keywords: spaceflight, transcriptomics, mitochondria, cross-species meta-analysis, NASA OSDR, orthology

1 Introduction

Spaceflight exposes organisms to a unique combination of environmental stressors including microgravity, cosmic radiation, altered atmospheric composition, and circadian disruption. These conditions induce measurable transcriptomic changes across diverse organisms, from yeast to humans [**TODO: citation needed**]. The NASA Open Science Data Repository (OSDR) has accumulated hundreds of spaceflight omics datasets through the GeneLab consortium, providing an unprecedented opportunity for cross-species comparative analysis [1].

Individual species studies have identified recurrent themes in the spaceflight transcriptomic response, including altered energy metabolism, oxidative stress, immune dysregulation, and cytoskeletal remodeling [**TODO: citation needed**]. However, most analyses have been conducted within single species, making it difficult to distinguish conserved fundamental responses from species-specific adaptations. A rigorous cross-species comparison requires orthology-based integration, which maps genes across species to their common ancestral equivalents.

Subcellular localization provides a functional organizing principle for interpreting transcriptomic changes. Gene Ontology Cellular Component (GOCC) enrichment can identify which organelles are coordinately regulated, and cross-species conservation of these organelle-level responses would suggest deeply rooted cellular mechanisms. Mitochondria are of particular interest given their central role in energy metabolism, oxidative stress response, and apoptosis, all of which are affected by spaceflight [**TODO: citation needed**].

Here we present a cross-species meta-analysis of 22 spaceflight transcriptomic datasets spanning six eukaryotic species. We build a human-anchored orthology matrix using OrthoDB v12, perform subcellular location enrichment, and apply Fisher's combined probability test to identify conserved organelle-level responses. We further visualize conserved metabolic pathways with per-species expression overlays and map cofactor dependencies. Our analysis reveals that mitochondrial gene suppression is the most robustly conserved subcellular response to spaceflight across a billion years of eukaryotic evolution.

2 Results

2.1 Dataset overview and differential expression

We queried the NASA OSDR Biological Data API for RNA-seq and microarray datasets with spaceflight versus ground control contrasts across six species. The final selection comprised 22 datasets: 7 mouse, 4 human, 3 Arabidopsis, 3 fly, 4 worm (microarray), and 1 yeast (microarray) (Fig. 1A). RNA-seq datasets used GeneLab Research Community Portal (RCP) pre-computed DESeq2 tables; microarray datasets used limma. For two datasets without pre-computed tables (OSD-96, OSD-35), we performed de novo differential expression analysis.

We applied tiered DEG thresholds to accommodate dataset-specific statistical power: standard ($p_{\text{adj}} < 0.05$, $|\log2\text{FC}| > 1$) for 19 datasets, relaxed ($p_{\text{adj}} < 0.10$, $|\log2\text{FC}| > 0.5$) for two low-power datasets, and nominal ($p < 0.05$, $|\log2\text{FC}| > 0.5$)

for one dataset without replicates. This yielded 20,333 DEGs total, with substantial variation across species: Arabidopsis (9,930), *C. elegans* (5,022), mouse (2,103), fly (1,881), human (1,366), and yeast (31) (Fig. 1B). The low yeast count reflects the single microarray dataset with a relaxed threshold. Across all species, 10,931 DEGs were upregulated and 9,402 downregulated in spaceflight relative to ground controls.

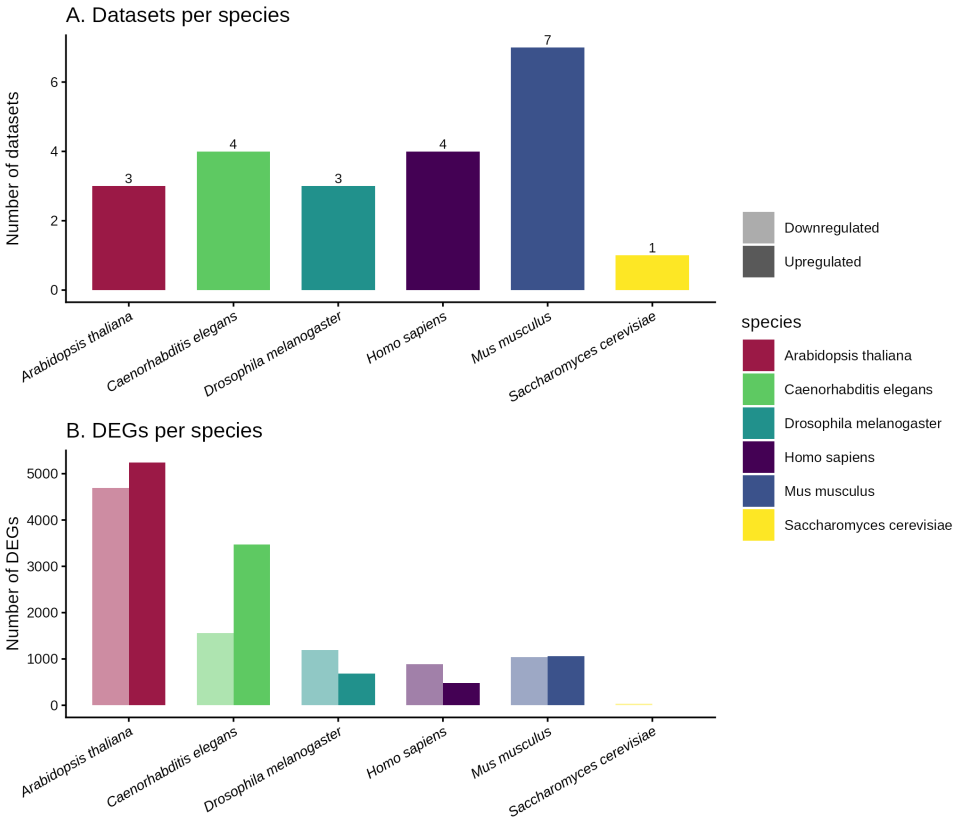


Fig. 1 Study overview. (A) Number of OSDR datasets per species included in the analysis. (B) Number of differentially expressed genes (DEGs) per species, separated by direction (upregulated vs. downregulated in spaceflight relative to ground control). Species colors are fixed globally across all figures using a viridis-based colorblind-friendly palette.

2.2 Human-anchored orthology mapping

To enable cross-species comparison, we built a human-anchored orthology matrix using OrthoDB v12 as the primary resource, supplemented by babelgene for human-to-mouse, fly, worm, and yeast mappings. Arabidopsis orthology was mapped via OrthoDB Eukaryota-level ortholog groups, as babelgene does not support plant

species. The unified matrix comprises 18,030 human genes with at least one non-human ortholog, of which 651 have orthologs in all five other species, 3,446 in at least four, and 8,505 in at least three (Fig. 2).

DEG-to-ortholog mapping rates varied substantially across species, reflecting both evolutionary distance and gene identifier compatibility: human 100%, mouse 70.7%, yeast 64.5%, fly 28.6%, worm 18.8%, and Arabidopsis 3.2%. The low Arabidopsis rate reflects the large plant-specific gene complement and the reliance on OrthoDB rather than babelgene for plant orthology. Despite this, 352 human orthologs were identified as DEGs in two or more species, and 33 in three or more, providing a robust foundation for conservation analysis.

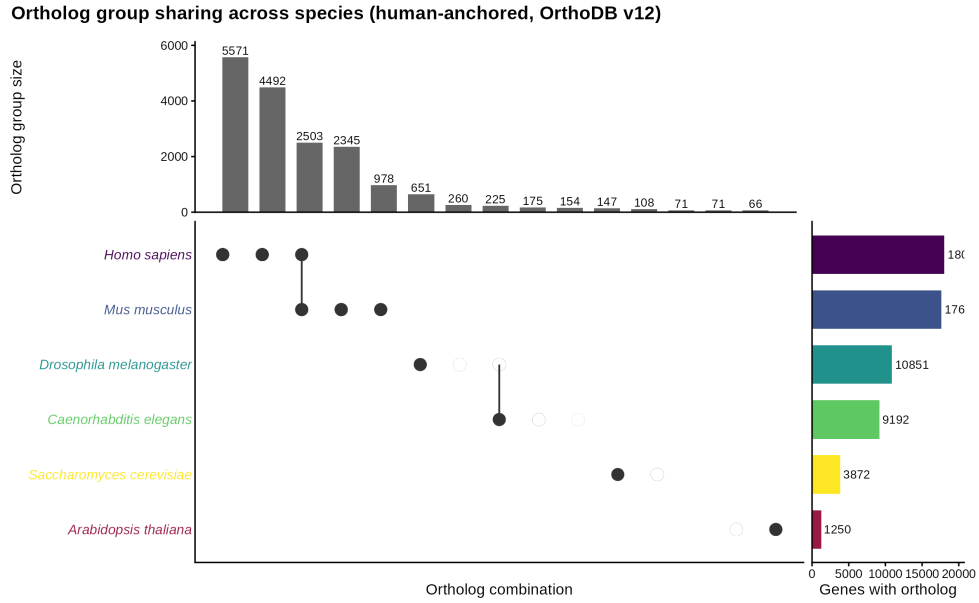


Fig. 2 Ortholog group sharing across species. UpSet plot showing the number of human genes with orthologs in each combination of the five non-human species (OrthoDB v12 + babelgene). Human is the anchor species (present in all groups). 651 genes have orthologs in all five non-human species; 8,505 in at least three.

2.3 Species-specific transcriptional responses

Volcano plots for representative datasets (the dataset with the most DEGs per species) reveal both shared and species-specific patterns (Fig. 3). All species show a predominance of downregulated genes in the most affected datasets, particularly in fly (OSD-207, 1,735 DEGs) and human (OSD-684, 971 DEGs). Arabidopsis (OSD-217) shows an unusually high DEG count (9,069), likely reflecting the strong sensitivity of plant transcriptomes to the combined stressors of spaceflight including altered gravity and light conditions.

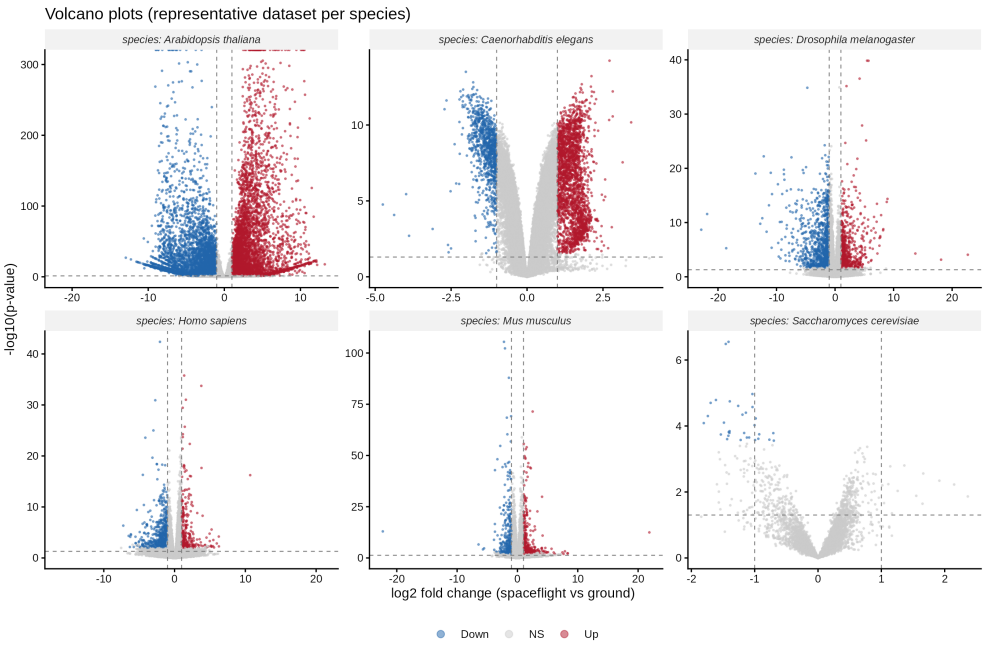


Fig. 3 Volcano plots of representative datasets per species. Each panel shows the dataset with the most DEGs for that species. Dashed lines indicate $p = 0.05$ (horizontal) and $|\log2FC| = 1$ (vertical). Red = upregulated in spaceflight, blue = downregulated, grey = not significant. OSD IDs: Arabidopsis OSD-217, *C. elegans* OSD-112, Fly OSD-207, Human OSD-684, Mouse OSD-104, Yeast OSD-62.

2.4 Subcellular location enrichment reveals organelle-specific responses

We performed GOCC enrichment analysis per species for up- and down-regulated DEGs separately using clusterProfiler, yielding 200 enriched terms. We manually classified all terms into 16 organelle categories. Seven categories were enriched in three or more species: cell wall/extracellular matrix, plasma membrane, extracellular, cytoskeleton, nucleus, proteasome, and ribosome (Fig. 4).

The signed enrichment score (negative for downregulated, positive for upregulated) reveals a striking pattern: mitochondrion, membrane transport, cytoskeleton, and endoplasmic reticulum show predominantly negative scores (downregulated enrichment) across species, while peroxisome, lipid particle, Golgi apparatus, and chloroplast show positive scores (upregulated enrichment). The nucleus shows a mixed response with both up- and down-regulated components, consistent with its diverse functional roles.

2.5 Conserved mitochondrial suppression across species

To formally test for cross-species conservation, we applied Fisher's combined probability test to the per-species p-values of each orthologous gene within each organelle

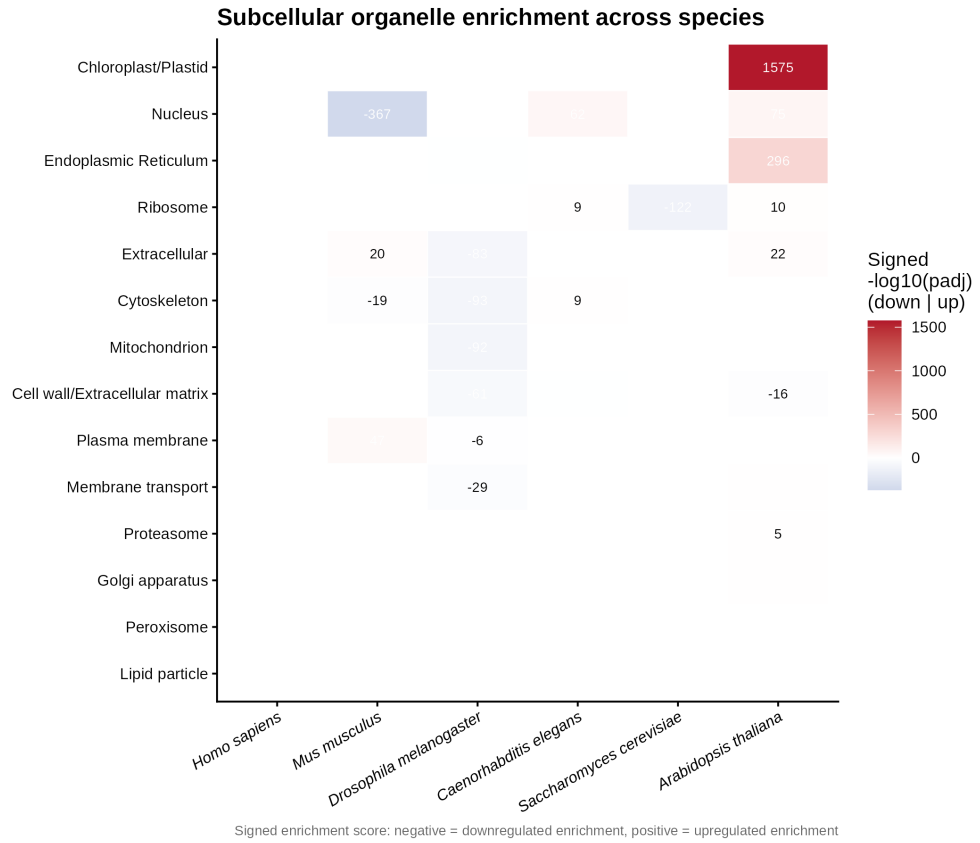


Fig. 4 Subcellular organelle enrichment across species. Heatmap shows signed enrichment scores (sum of $-\log_{10}$ adjusted p-value, signed by direction) for each organelle category and species. Blue = downregulated enrichment, red = upregulated enrichment. Organelles are ordered by total absolute enrichment signal.

category. Of 1,210 ortholog-organelle tests, 1,079 were significant after Benjamini-Hochberg FDR correction (FDR < 0.05). We defined a gene as 'conserved' if it was significant in at least three species with consistent direction ($\geq 80\%$ same direction).

The most striking finding is the conserved downregulation of mitochondrial genes: 49 orthologs show significant, direction-consistent suppression across at least three species (100% downregulated, minimum Fisher FDR $= 7.7 \times 10^{-68}$) (Table 1, Fig. 5). The top genes include components of all major oxidative phosphorylation complexes and the TCA cycle: IDH3B (isocitrate dehydrogenase, 5 species, $p = 7.7 \times 10^{-68}$), CYC1 (cytochrome c1, Complex III, 6 species, $p = 3.1 \times 10^{-39}$), COX6A2 and COX6B2 (Complex IV, 5 species each), COX5B (Complex IV, 5 species), ATP5F1A and ATP5PD (ATP synthase, 5 species each), and IDH3A/IDH3G (TCA cycle, 5 species each). Notably, two genes (CYC1 and PDHB) are conserved downregulated across all six species, and GPD2 across all six.

Table 1 Top conserved downregulated mitochondrial genes across species.

Symbol	Function	Complex/Pathway	Species (n)	Fisher FDR
IDH3B	Isocitrate dehydrogenase 3B	TCA cycle	5	7.7×10^{-68}
CYC1	Cytochrome c1	Complex III	6	3.1×10^{-39}
COX6A2	Cytochrome c oxidase subunit 6A2	Complex IV	5	1.8×10^{-27}
COX6B2	Cytochrome c oxidase subunit 6B2	Complex IV	5	1.8×10^{-27}
COX5B	Cytochrome c oxidase subunit 5B	Complex IV	5	5.6×10^{-27}
ATP5F1A	ATP synthase F1 alpha	ATP synthase	5	9.8×10^{-25}
IDH3G	Isocitrate dehydrogenase 3G	TCA cycle	5	4.8×10^{-24}
IDH3A	Isocitrate dehydrogenase 3A	TCA cycle	5	1.3×10^{-22}
PDHB	Pyruvate dehydrogenase E1 beta	Pyruvate dehydrogenase	6	6.2×10^{-20}
ATP5PD	ATP synthase peripheral stalk	ATP synthase	5	1.0×10^{-18}
GPD2	Glycerol-3-phosphate dehydrogenase 2	Mitochondrial shuttle	6	7.5×10^{-18}
SLC25A6	ADP/ATP translocase 3	Mitochondrial transport	4	5.4×10^{-18}
NDUFA10	NADH dehydrogenase 1 alpha 10	Complex I	4	7.5×10^{-18}

In contrast to the mitochondrial suppression, several organelle categories show conserved upregulation: peroxisome (6 genes, 100% up, $p = 8.4 \times 10^{-105}$), lipid particle (5 genes, 100% up, $p = 1.2 \times 10^{-21}$), and Golgi apparatus (2 genes, 100% up, $p = 3.3 \times 10^{-130}$). The proteasome shows a mixed response (5 conserved genes: 4 down, 1 up), and the ribosome is nearly balanced (31 conserved: 16 down, 15 up). The full conservation summary is provided in Table S7.

2.6 Pathway-level visualization reveals cofactor-dependent divergence

To examine the conserved mitochondrial suppression at pathway resolution, we visualized four KEGG pathways (oxidative phosphorylation, proteasome, ribosome, TCA cycle) using ggkegg, overlaying per-species mean log2FC values onto gene nodes (Fig. 6). This reveals that while the overall mitochondrial suppression is conserved, individual complex components show species-specific patterns.

In oxidative phosphorylation (hsa00190), *Drosophila* shows the strongest downregulation across Complex I, III, IV, and V components, while human and mouse show more modest changes with some upregulation of specific subunits. *C. elegans* and Arabidopsis show intermediate patterns. The TCA cycle (hsa00020) shows more uniform downregulation across species, particularly for the isocitrate dehydrogenase complex (IDH3A, IDH3B, IDH3G), consistent with the conservation meta-analysis.

2.7 Cofactor-dependent enzymes show species-divergent responses

We mapped 14 enzyme cofactors to human genes via the KEGG compound-reaction-enzyme-gene chain, yielding 498 unique genes across 12 cofactors (Fe-S cluster and Zn^{2+} returned no genes as KEGG tracks these as prosthetic groups rather than reaction substrates). We then examined the expression of cofactor-dependent enzymes across species and pathways (Fig. 7).

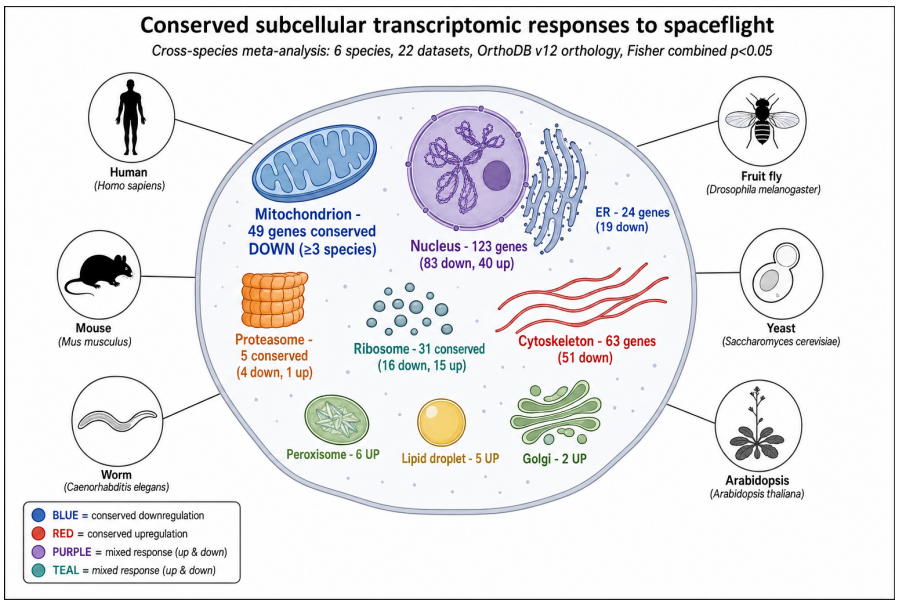


Fig. 5 Conserved subcellular transcriptomic responses to spaceflight. Schematic of a eukaryotic cell showing organelles with conserved transcriptional responses across species. Blue indicates conserved downregulation, red indicates conserved upregulation, and purple/teal indicate mixed responses. The mitochondrion (49 genes, 100% downregulated across ≥ 3 species) is the most robustly conserved response. Numbers indicate conserved gene counts.

NAD^+/NADH-dependent enzymes, which dominate oxidative phosphorylation, show species-divergent responses: downregulation in *Drosophila* (mean log2FC $= -0.18$) but upregulation in human (0.14) and mouse (0.17). In contrast, CoA-dependent enzymes in the TCA cycle show more variable but generally modest changes. TPP-dependent enzymes (pyruvate dehydrogenase complex) show consistent downregulation in *Drosophila* and *C. elegans*. Heme-dependent enzymes (cytochrome components) show downregulation in *Drosophila* and human but slight upregulation in yeast. These cofactor-level patterns suggest that while the overall mitochondrial suppression is conserved, the specific enzymatic routes affected vary by species, potentially reflecting differences in metabolic strategy or redox buffering capacity.

3 Discussion

Our cross-species meta-analysis reveals that mitochondrial gene suppression is the most robustly conserved subcellular transcriptomic response to spaceflight, with 49 genes showing significant, direction-consistent downregulation across at least three of six eukaryotic species spanning approximately one billion years of evolution. This finding extends previous single-species observations of mitochondrial dysregulation in spaceflight [**TODO: citation needed**] to a conserved, multi-kingdom response.

The specific genes identified paint a coherent biological picture: the conserved suppression targets all five oxidative phosphorylation complexes (I–V) and key TCA

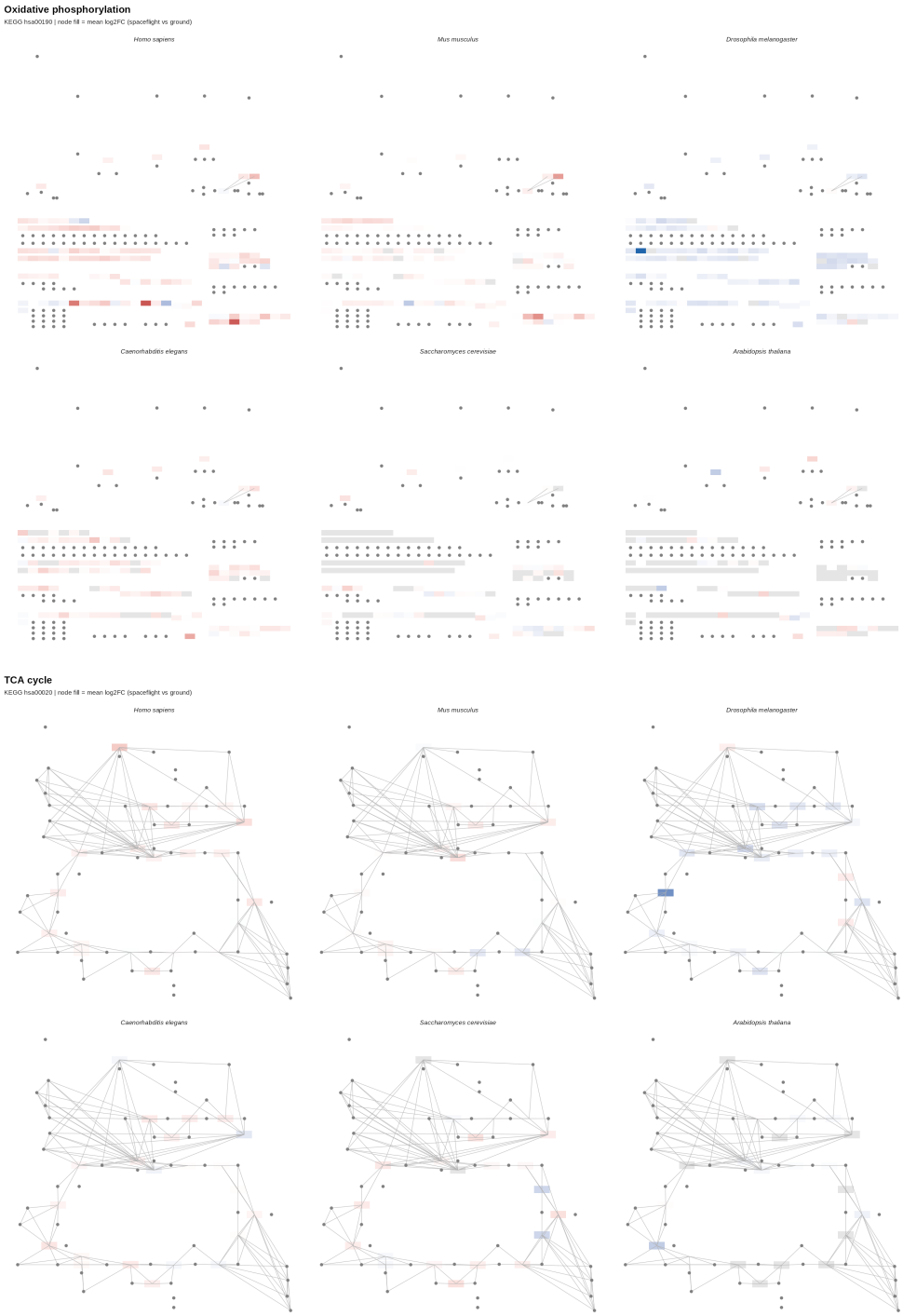


Fig. 6 KEGG pathway diagrams with per-species log2FC overlay. Top: oxidative phosphorylation (hsa00190). Bottom: TCA cycle (hsa00020). Each panel shows the KEGG pathway for one species, with gene nodes colored by mean log2FC (blue = downregulated, red = upregulated, grey = no data). Diverging scale: −2 to +2. Proteasome (hsa03050) and ribosome (hsa03010) are provided as supplementary figures.

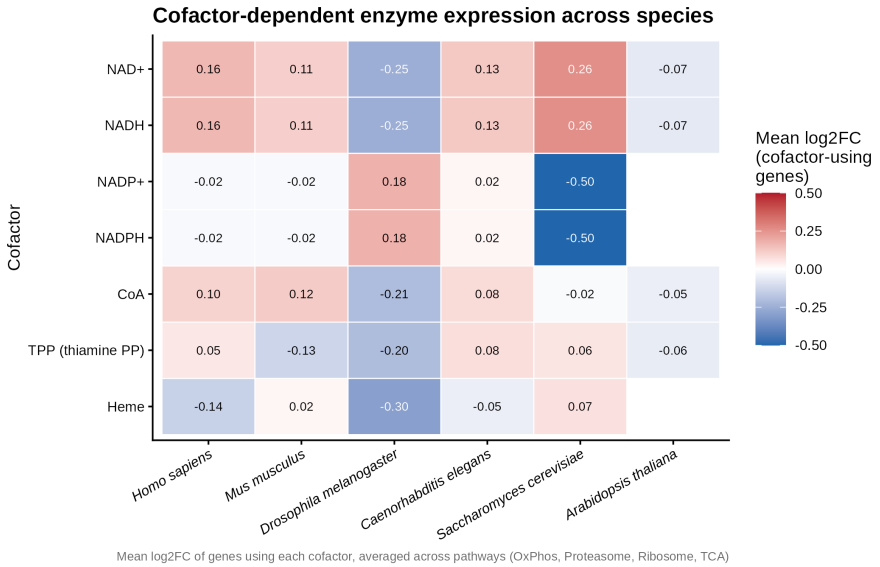


Fig. 7 Cofactor-dependent enzyme expression across species. Heatmap shows mean log2FC of genes using each cofactor, averaged across the four analyzed pathways. Blue = downregulated, red = upregulated. NAD^+/NADH enzymes show the most species-divergent pattern; CoA and TPP enzymes show more conserved responses.

cycle enzymes, particularly the isocitrate dehydrogenase complex (IDH3A, IDH3B, IDH3G). This suggests a coordinated reduction in oxidative capacity rather than a stochastic effect. The conservation of this response from yeast to human implies a fundamental cellular mechanism, potentially related to reduced energy demand in microgravity, oxidative stress mitigation (reduced electron transport reduces reactive oxygen species production), or a shift toward glycolytic metabolism.

The contrast between conserved mitochondrial suppression and conserved peroxisome and lipid droplet upregulation is intriguing. Peroxisomes are central to fatty acid beta-oxidation and hydrogen peroxide metabolism, and their upregulation may reflect compensatory responses to altered lipid metabolism or oxidative stress. Lipid droplet accumulation has been observed in multiple spaceflight studies and may relate to altered lipid handling in microgravity [2] [**TODO: citation needed**]. The coordinated upregulation of Golgi apparatus genes suggests enhanced secretory pathway activity, potentially related to extracellular matrix remodeling or stress signaling.

Our cofactor analysis reveals an important nuance: while the overall mitochondrial suppression is conserved, the specific enzymatic routes affected vary by species. NAD^+/NADH-dependent enzymes show divergent responses (down in fly, up in human and mouse), while CoA-dependent and TPP-dependent enzymes show more conserved patterns. This suggests that species may employ different metabolic strategies in response to spaceflight, with the conserved feature being a reduction in specific mitochondrial functions rather than a uniform suppression of all oxidative metabolism.

Several limitations should be noted. First, the datasets span different tissues, durations, and experimental designs, introducing biological and technical heterogeneity.

We addressed this by using tissue as a covariate where possible and applying tiered DEG thresholds, but residual heterogeneity remains. Second, orthology mapping rates vary substantially across species, with Arabidopsis particularly affected (3.2% DEG mapping rate) due to the plant-specific gene complement and reliance on OrthoDB rather than babelgene. Third, the yeast dataset is a single microarray study with only 31 DEGs, limiting statistical power for this species. Fourth, our conservation criterion requires significance in at least three species, which may miss genuinely conserved responses that fall below significance thresholds in individual datasets. Finally, transcriptomic changes do not necessarily reflect protein-level or functional changes, and the relationship between mRNA suppression and actual mitochondrial function in spaceflight requires further validation.

Despite these limitations, the robustness of the mitochondrial suppression signal, with Fisher combined p-values as low as 10^{-68} and consistent direction across diverse species, provides strong evidence for a conserved cellular response. The identification of specific targets (IDH3B, CYC1, COX5B, ATP5F1A) provides candidate biomarkers for monitoring mitochondrial responses to spaceflight and potential targets for countermeasure development. Future work should integrate proteomic and metabolomic data to validate the functional consequences of these transcriptomic changes and explore whether pharmacological mitigation of mitochondrial suppression is feasible and beneficial.

4 Methods

4.1 Data acquisition

We queried the NASA OSDR Biological Data API (`visualization.osdr.nasa.gov/biodata/api/v2/query/`) to enumerate RNA-seq and microarray datasets with spaceflight versus ground control contrasts for six species. The final selection comprised 22 datasets: OSD-37, 47, 48, 62, 96, 104, 105, 112, 113, 120, 207, 217, 242, 258, 323, 347, 35, 379, 421, 684, 863. Processed data (DE tables, count matrices, sample tables) were downloaded via the OSDR geode-py web service. Tissue metadata was extracted from the `study.characteristics.material type` field.

4.2 Differential expression analysis

For 19 datasets, we used GeneLab RCP pre-computed DE tables (DESeq2 [3] for RNA-seq, limma [4] for microarray). For OSD-96 (*Drosophila* RSEM counts), we ran DESeq2 with design $\sim$ tissue + condition. For OSD-35 (*C. elegans* microarray, no replicates), we ran limma with nominal thresholds. We applied tiered DEG thresholds: standard ($p_{\text{adj}} < 0.05$, $|\text{log2FC}| > 1$) for 19 datasets, relaxed ($p_{\text{adj}} < 0.10$, $|\text{log2FC}| > 0.5$) for two datasets, and nominal ($p < 0.05$, $|\text{log2FC}| > 0.5$) for one dataset. Direction was normalized so positive log2FC = upregulated in spaceflight.

4.3 Orthology mapping

We built a human-anchored orthology matrix using OrthoDB v12 [5] (`data.orthodb.org/v12/`) as the primary resource, supplemented by babelgene [6] for human-to-mouse, fly, worm, and yeast mappings. Arabidopsis TAIR IDs were mapped to human orthologs via OrthoDB Eukaryota-level ortholog groups. The unified matrix comprises 18,030 human genes with at least one non-human ortholog.

4.4 Subcellular location enrichment

We ran clusterProfiler [7] `enrichGO` (ontology = CC, $p < 0.05$, Benjamini-Hochberg) per species for up- and down-regulated DEGs separately, using species-specific `org.db` packages. Enriched terms were manually classified into 16 organelle categories.

4.5 Cross-species conservation meta-analysis

For each organelle category, we applied Fisher's combined probability test [8] to per-species p-values for each orthologous gene, then adjusted across all 1,210 tests using Benjamini-Hochberg FDR [9]. A gene was considered conserved if Fisher FDR < 0.05, significant in ≥ 3 species, and direction-consistent ($\geq 80\%$ same direction).

4.6 Pathway visualization and cofactor mapping

KEGG [10] pathway-gene links were fetched via the KEGG REST API (`rest.kegg.jp`) for hsa00190, hsa03050, hsa03010, hsa00020. Cofactor-gene mappings were built via the compound-reaction-enzyme-gene chain. Pathway diagrams were rendered using ggkegg [11] with per-species log2FC overlays (diverging blue-white-red scale, limits -2 to $+2$); palettes use viridis [12].

4.7 AI use disclosure

Data analysis scripting, figure generation, and manuscript drafting were assisted by Biomni (Phylo). All statistical analyses, data interpretation, and scientific conclusions were conducted and verified by the author. This AI assistance is disclosed in accordance with npj Microgravity editorial policy.

Data availability. All raw and processed data are available from the NASA Open Science Data Repository (`osdr.nasa.gov`) using the OSD identifiers listed in Methods. The complete analysis code, orthology matrices, supplementary tables, and figures are deposited in a Zenodo repository (DOI to be assigned upon submission). OrthoDB v12 data are available at `data.orthodb.org/v12/`. KEGG pathway data are available at `kegg.jp`.

Code availability. All analysis scripts (R) are available in the project repository under the `scripts/` directory. The analysis used R 4.3 with DESeq2, limma, clusterProfiler, ggkegg, babelgene, and OrthoDB v12 bulk files. Full software requirements are listed in the repository README.

Acknowledgements. We thank the NASA GeneLab consortium and the Open Science Data Repository for providing open access to spaceflight omics data. This analysis was assisted by Biomni (Phylo).

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